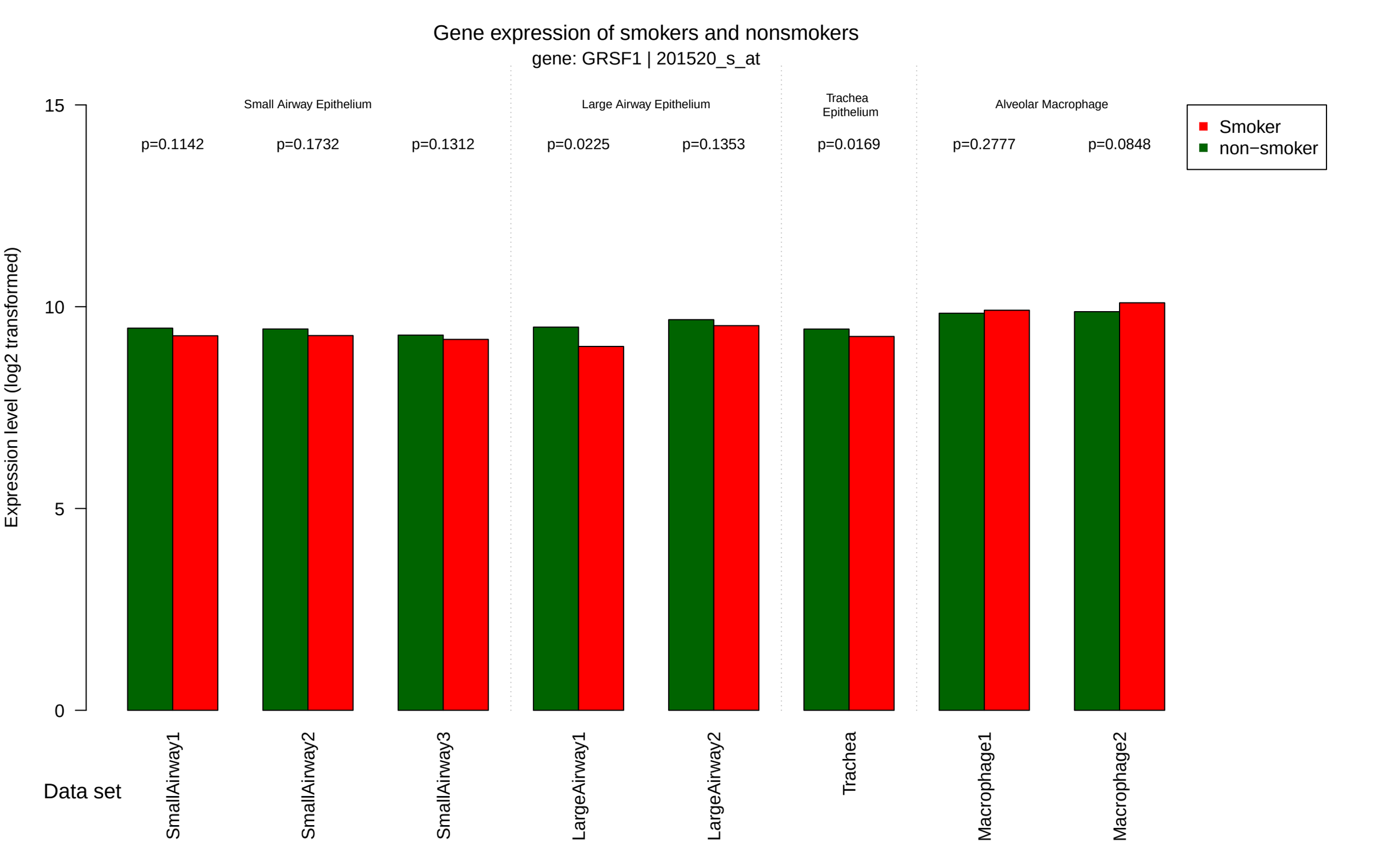
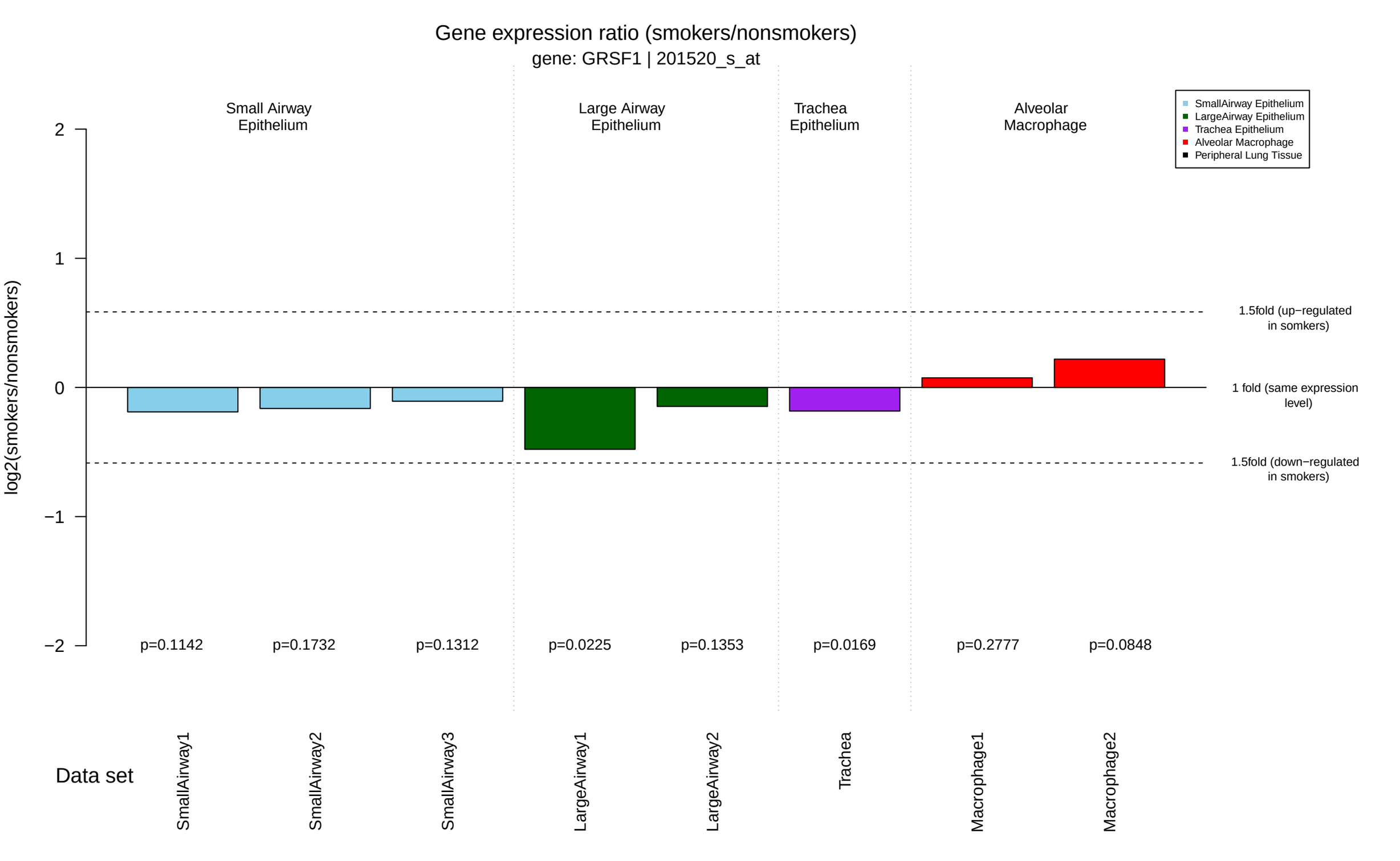
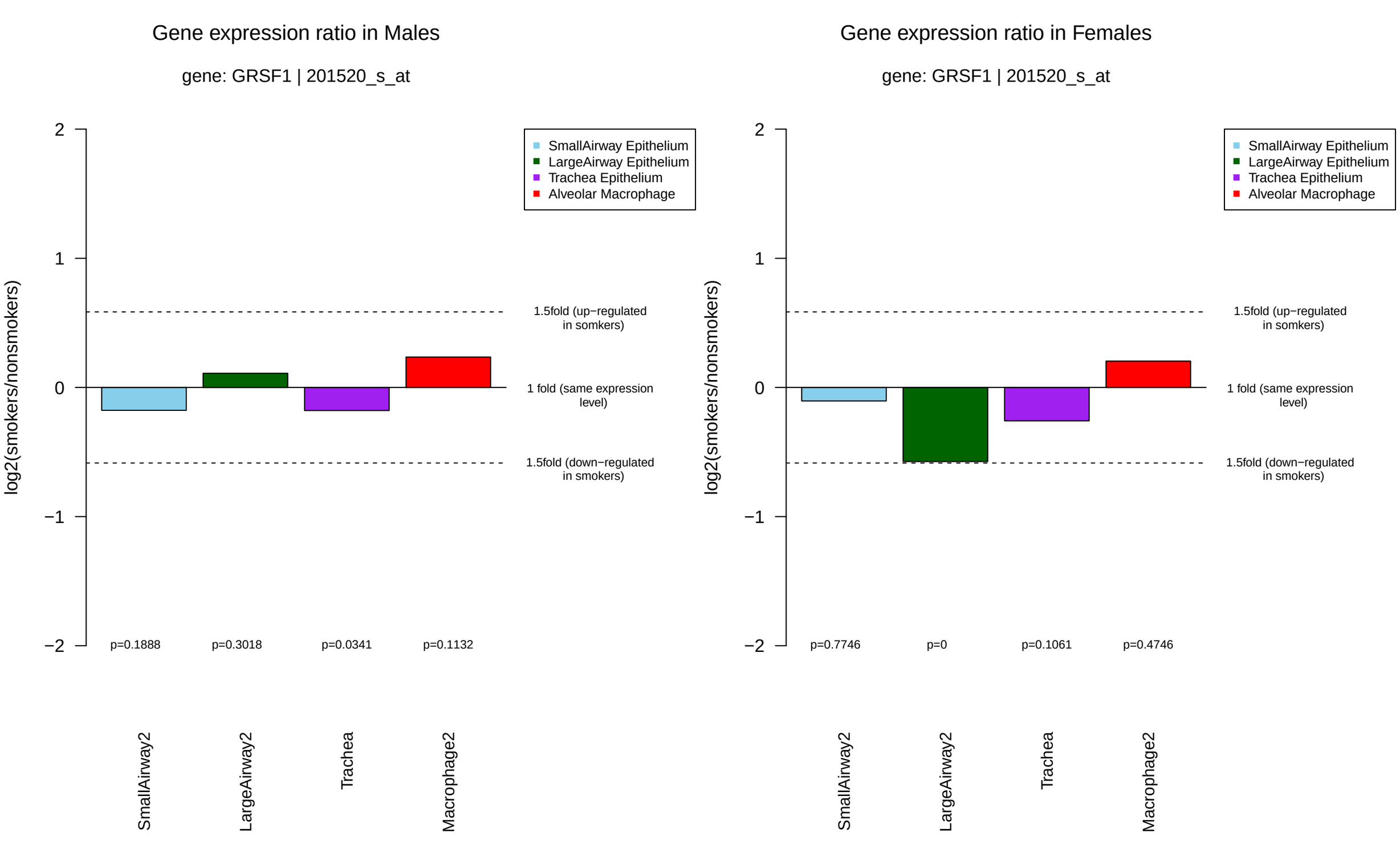
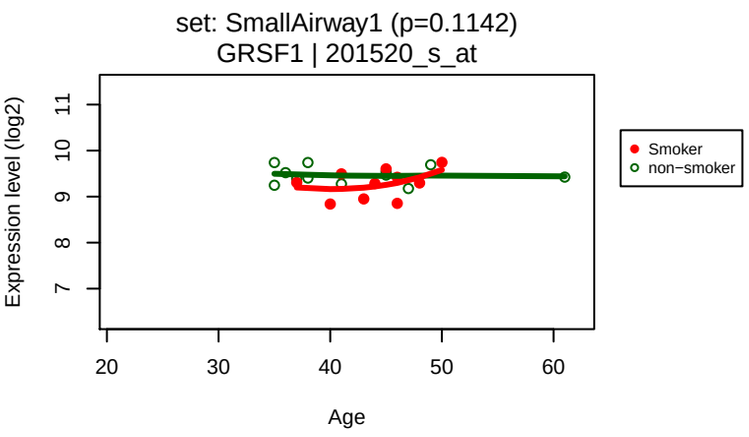
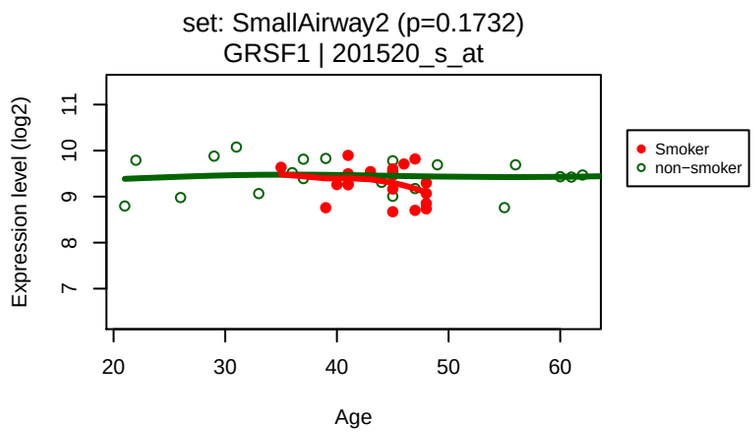
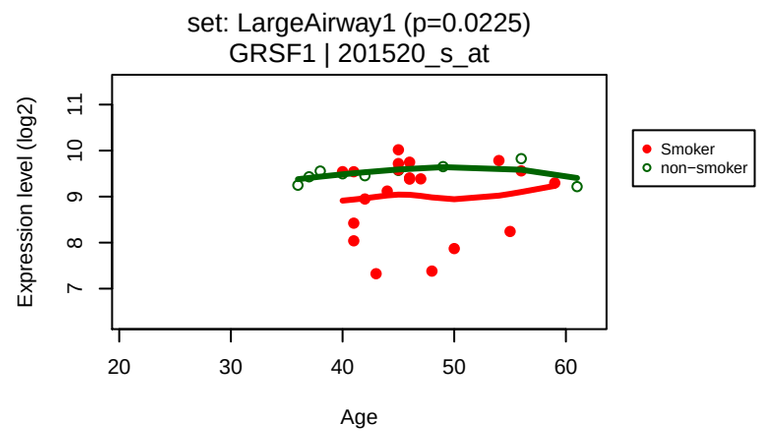

Gene expression of smokers and nonsmokers
gene: GRSF1 | 201520_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1142
p=0.1732
p=0.1312
p=0.0225
p=0.1353
p=0.0169
p=0.2777
p=0.0848
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

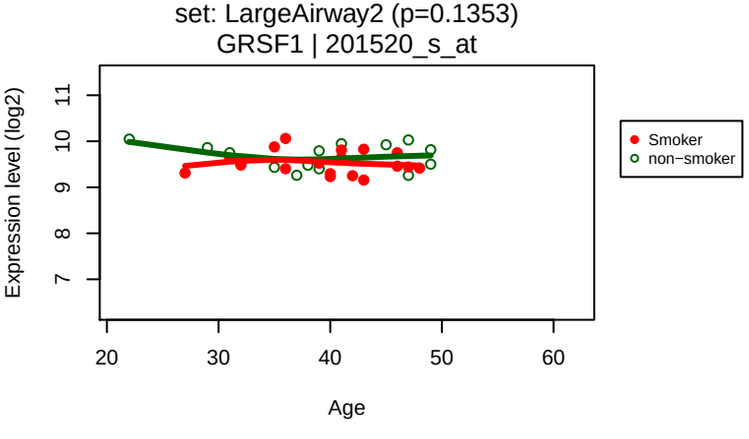

Gene expression ratio (smokers/nonsmokers)
gene: GRSF1 | 201520_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1142
p=0.1732
p=0.1312
p=0.0225
p=0.1353
p=0.0169
p=0.2777
p=0.0848
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

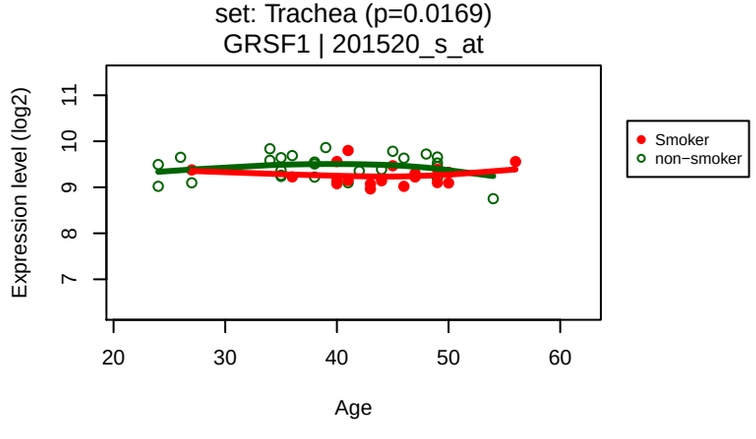

Gene expression ratio in Males
gene: GRSF1 | 201520_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1888
p=0.3018
p=0.0341
p=0.1132
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRSF1 | 201520_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7746
p=0
p=0.1061
p=0.4746
SmallAirway2
LargeAirway2
Trachea
Macrophage2

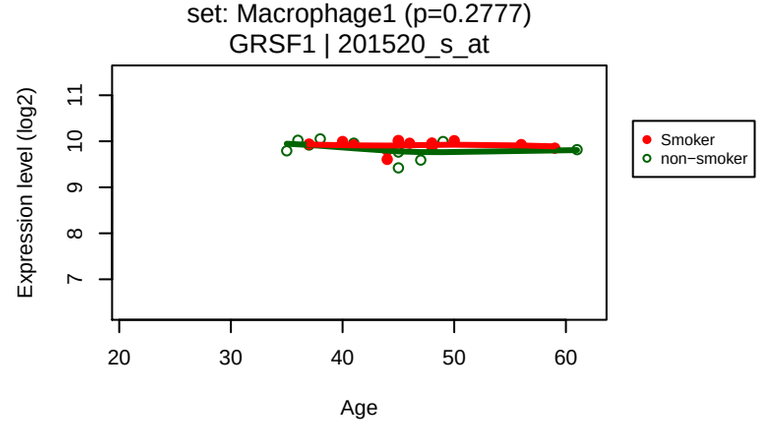

set: SmallAirway1 (p=0.1142)
GRSF1 | 201520_s_at
Expression level (log2)
Age
Smoker
non−smoker

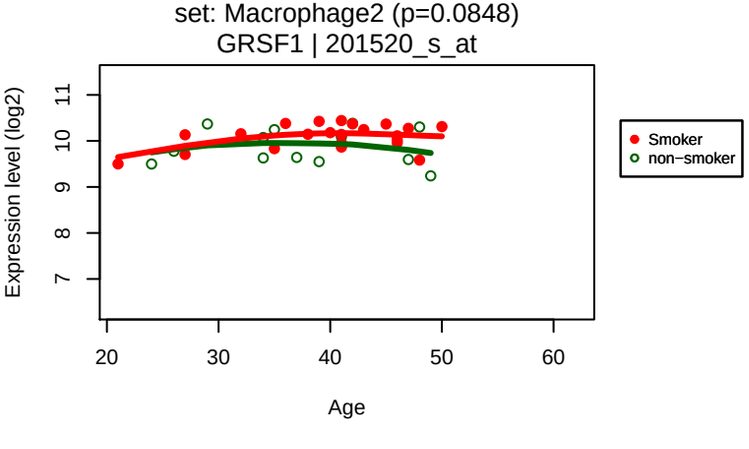

set: SmallAirway2 (p=0.1732)
GRSF1 | 201520_s_at
Expression level (log2)
Age
Smoker
non−smoker

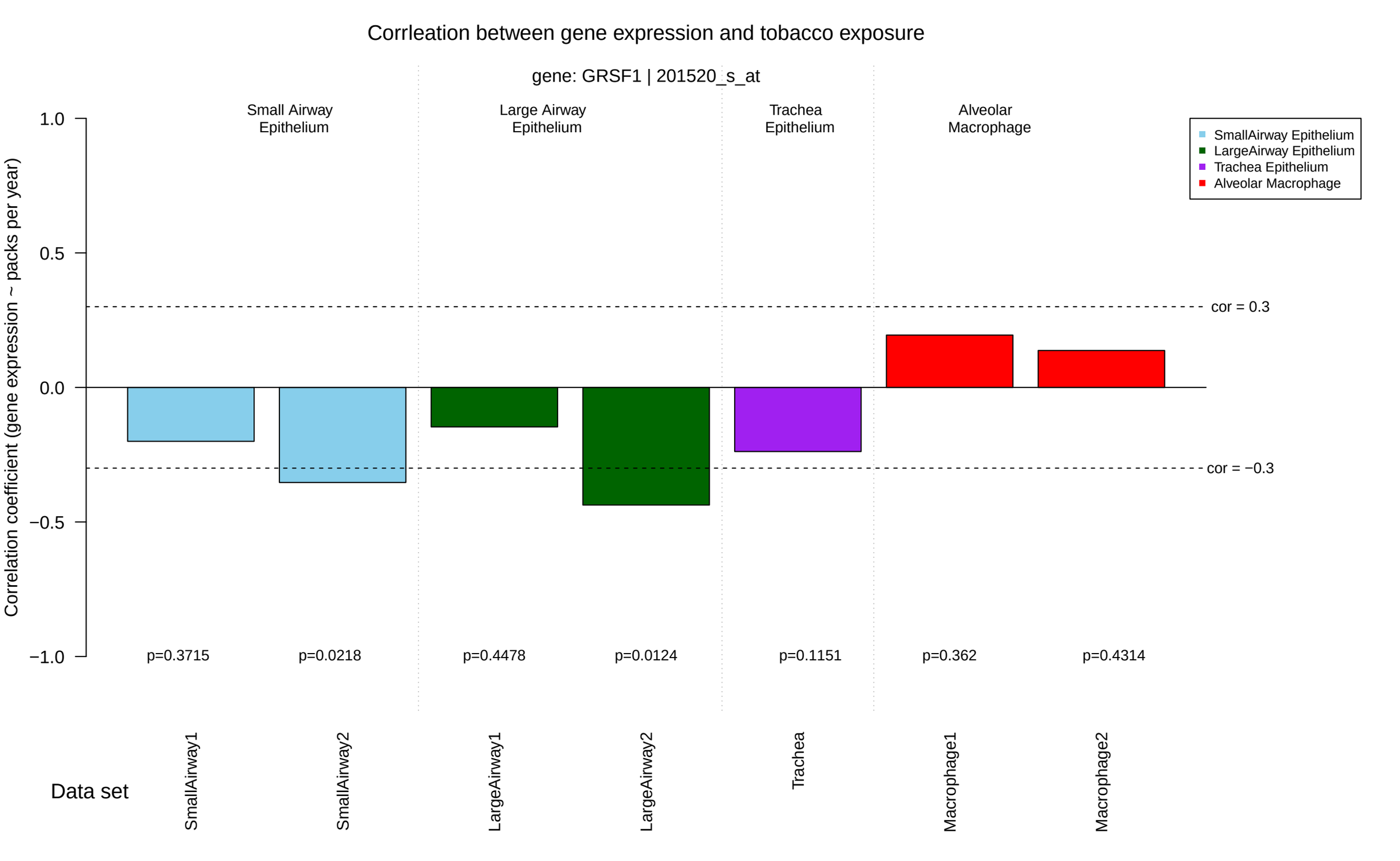

set: LargeAirway1 (p=0.0225)
GRSF1 | 201520_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1353)
GRSF1 | 201520_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0169)
GRSF1 | 201520_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2777)
GRSF1 | 201520_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0848)
GRSF1 | 201520_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRSF1 | 201520_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3715
p=0.0218
p=0.4478
p=0.0124
p=0.1151
p=0.362
p=0.4314
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2